Padraig Lavery

Software Engineer – Application Security & Compliance

Belfast, Northern Ireland | LinkedIn | GitHub

PROFILE

Software engineer with 4+ years building and securing high-volume banking systems for UK and Middle East clients, using Node.js, .NET/C# and PostgreSQL on AWS and Kubernetes. Has delivered PCI DSS vulnerability remediation, audit logging for client security audits, and hardening of authentication and account services. Pairs security work with strong production reliability and clear documentation for other engineers.

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

Security and compliance

- Designed and implemented audit logging of staff access to customer accounts across UK and Middle East bank portals (agent, customer, timestamp, session, IP) to meet client security and compliance audit requirements.
- Hardened authentication and account services: removed stack traces and SQL details from error responses and standardised input validation across partner microservices to stop malformed requests triggering database errors.
- Remediated critical Snyk-flagged vulnerabilities for PCI DSS compliance, alongside planned AWS SDK v3 and Node 22 Lambda runtime upgrades across repositories and bank deployments.
- Built a secure e-code resend flow with audit-logged resends and multi-bank email routing.
- Audited 100+ S3 buckets, applying lifecycle policies and removing obsolete resources to reduce compliance risk.

Engineering and reliability

- Authored the AWS SDK for JavaScript v2-to-v3 migration guide across Lambdas, Fargate services and microservices.
- Eliminated recurring daily out-of-memory crashes in a transaction ingestion service by loading caches once per batch instead of per file.
- Resolved a customer-facing outage in a Kubernetes/Istio ingress setup and corrected the Helm templates behind it.
- Led a cross-team initiative for automated API testing, adding request validation, shared error handling, OpenAPI docs, unit tests and stable GitHub Actions CI.
- Authored runbooks and SOPs; mentored four new starters and run the team's retrospectives.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Developed a cloud-cost API using Go, Docker, Kubernetes and Git.

PROJECTS

AI-Based Credit Card Fraud Detection (final year project) | Angular, Flask, GraphQL, Docker, Kubernetes, AWS SageMaker

- End-to-end web app flagging irregular credit card transactions, with a fraud-detection model trained in R and Python and served via SageMaker.

SKILLS

- **Security:** PCI DSS remediation, Snyk, OWASP, audit logging, input validation, secure error handling
- **Languages:** JavaScript (Node.js), C#, Python, Go, SQL
- **Back end and data:** Node.js, .NET Core, microservices, OpenAPI/Swagger, PostgreSQL, MongoDB
- **Cloud and infrastructure:** AWS (Lambda, Fargate, S3, EventBridge), Kubernetes, Helm, Istio, Docker, Terraform, GitHub Actions

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022
- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026